
Lecture Notes

on **Vector Algebra**
by Arnaja Mitra

1 Vectors

1.1 Definition

A **vector** is an ordered list of numbers (called **components** or **entries**).

The **dimension** of a vector is the number of entries it has. Formally, an n -dimensional vector is an element of $\mathbb{R}^n$.

1.2 Row Vector

An n -dimensional **row vector** is written horizontally as

$$\vec{u} = (u_1, u_2, u_3, \dots, u_n),$$

where $u_i \in \mathbb{R}$ are the entries.

✚ **Example:**

$$\vec{u} = (2, -1, 4, 7).$$

1.3 Column Vector

An m -dimensional **column vector** is written vertically as

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{pmatrix},$$

where $v_i \in \mathbb{R}$ are the entries.

✚ **Example:**

$$\vec{v} = \begin{pmatrix} 3 \\ 0 \\ -5 \end{pmatrix}.$$

2 Vector Operations

2.1 Definition

Given two vectors $\vec{u} = (u_1, u_2, \dots, u_n)$ and $\vec{v} = (v_1, v_2, \dots, v_n)$ in $\mathbb{R}^n$, we can define fundamental operations, such as addition, and scalar multiplication.

2.2 Vector Addition

The sum of two n -dimensional vectors is obtained by adding their corresponding components:

$$\vec{u} + \vec{v} = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n).$$

☛ **Example:** If $\vec{u} = (1, 2, 3)$ and $\vec{v} = (4, -1, 2)$, then

$$\vec{u} + \vec{v} = (5, 1, 5).$$

2.3 Scalar Multiplication

Multiplying a vector by a scalar $c \in \mathbb{R}$ scales each component:

$$c\vec{u} = (cu_1, cu_2, \dots, cu_n).$$

☛ **Example:** If $c = 3$ and $\vec{u} = (1, 2, 3)$, then

$$3\vec{u} = (3, 6, 9).$$

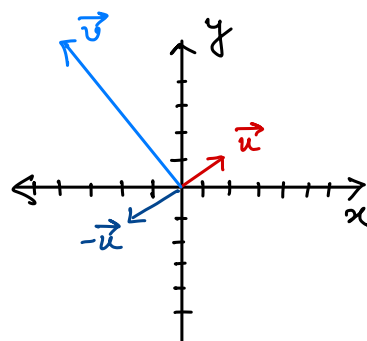
2.4 Length (Norm) of a Vector

The length (or magnitude) of $\vec{u}$ is

$$\|\vec{u}\| = \sqrt{u_1^2 + u_2^2 + \dots + u_n^2}.$$

☛ **Example:** If $\vec{u} = (3, 4)$, then

$$\|\vec{u}\| = \sqrt{3^2 + 4^2} = 5.$$



$$\vec{u} = (2, 1)$$
$$\vec{v} = (-3, 5)$$

2.5 Zero Vector

The vector in $\mathbb{R}^n$ whose entries are all zero is called the **zero vector**, denoted by

$$\vec{0} = (0, 0, \dots, 0).$$

2.6 Unit Vector

A vector of length 1 is called a **unit vector**. Any nonzero vector $\vec{u}$ can be converted into a unit vector by dividing by its length:

$$\hat{u} = \frac{\vec{u}}{\|\vec{u}\|}.$$

In particular for a 2-dim vector $\vec{u} = (u_1, u_2)$, the corresponding unit vector is given by $\frac{\vec{u}}{\|\vec{u}\|} = \left(\frac{u_1}{\sqrt{u_1^2 + u_2^2}}, \frac{u_2}{\sqrt{u_1^2 + u_2^2}} \right) = \vec{m} \text{ (say)}$
Check if unit or not?

$$\|\vec{m}\| = \sqrt{\frac{u_1^2}{(u_1^2 + u_2^2)} + \frac{u_2^2}{(u_1^2 + u_2^2)}} = \sqrt{\frac{u_1^2 + u_2^2}{u_1^2 + u_2^2}} = \sqrt{1} = 1. \text{ (Proved)}$$

Example: Construct a vector that represents the composition of a coastal wetlands landscape (which includes submerged, saturated but not submerged, and dry). Given that at a fixed time t , 65% of the wetland was submerged, 30% saturated, and 5% dry:

1. Represent this information as a vector.
2. If the total area of the landscape is 160 hectares, how many hectares are in each of the three classes?
3. Also find the corresponding percentage of each of the classes.

1) let $\vec{u}$ denotes the vector consisting the composition of the coastal wetland, and is given by

$$\vec{u} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}, \quad \begin{array}{lll} u_1 - \text{denote the submerged area} & & \\ u_2 - \text{" " " saturated} & & \\ u_3 - \text{" " " dry} & & \end{array}$$

So according to the given information

$$\vec{u} = \begin{pmatrix} 0.65 \\ 0.3 \\ 0.05 \end{pmatrix}.$$

$$65\% = \frac{65}{100} = 0.65$$

2) The distribution of the 160 hectares land among the three types of areas is given by

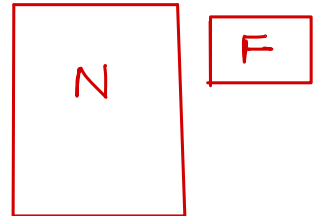
$$\vec{v} = 160\vec{u} = 160 \begin{pmatrix} 0.65 \\ 0.3 \\ 0.05 \end{pmatrix} = \begin{pmatrix} 104 \\ 48 \\ 8 \end{pmatrix}.$$

3) The corresponding percentage is given by

$$= \frac{1}{(104 + 48 + 8)} \begin{pmatrix} 104 \\ 48 \\ 8 \end{pmatrix} \times 100\% = \frac{1}{160} \begin{pmatrix} 104 \\ 48 \\ 8 \end{pmatrix} \times 100\% = \begin{pmatrix} 65\% \\ 30\% \\ 5\% \end{pmatrix}.$$

✎ **Example:** Suppose we are managing a national park. For the entire park, the flora composition is as follows: 8% annual plants, 23% perennial plants and grasses, 26% shrubs, 24% softwood trees and pines, 19% hardwood trees. The total area of the park is 1250 hectares. The park is now considering purchasing 300 hectares of farmland located next to the park, of which 50% is annual plants, 90% is perennial plants and grasses, 5% is softwood trees and pines. If the national park acquires this new land, what will be the new flora composition for the park?

let $\vec{u} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{pmatrix}$ denotes the flora composition for the national park.



& $\vec{w} = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \end{pmatrix}$ denotes the flora composition for the farmland.

Here the 1st, 2nd, 3rd, 4th & 5th components denote number of annual, perennial, shrubs, softwood and hardwood plants resp.

Given,

$$\vec{u} = \begin{pmatrix} 0.08 \\ 0.23 \\ 0.26 \\ 0.24 \\ 0.19 \end{pmatrix}, \quad \& \quad \vec{w} = \begin{pmatrix} 0.05 \\ 0.90 \\ 0 \\ 0.05 \\ 0 \end{pmatrix}.$$

So if we combine the national park and the farmland, then the required number of plants (all the 5 types)

$$\begin{aligned} 1250\vec{u} + 300\vec{w} &= \begin{pmatrix} 100 \\ 287.5 \\ 325 \\ 300 \\ 237.5 \end{pmatrix} + \begin{pmatrix} 15 \\ 270 \\ 0 \\ 15 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 115 \\ 557.5 \\ 325 \\ 315 \\ 237.5 \end{pmatrix} \end{aligned}$$

So the required composition (in percentage)

$$= \frac{1}{(1250+300)} \begin{pmatrix} 115 \\ 557.5 \\ 325 \\ 315 \\ 237.5 \end{pmatrix} \times 100\% \approx \begin{pmatrix} 7.42\% \\ 35.97\% \\ 20.97\% \\ 20.32\% \\ 15.32\% \end{pmatrix}.$$
